MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR,
(1 SHIELDED PAIR AND 8 WIRES UNDER A DOUBLE SHIELD)
M27072/76SACA*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-27072.

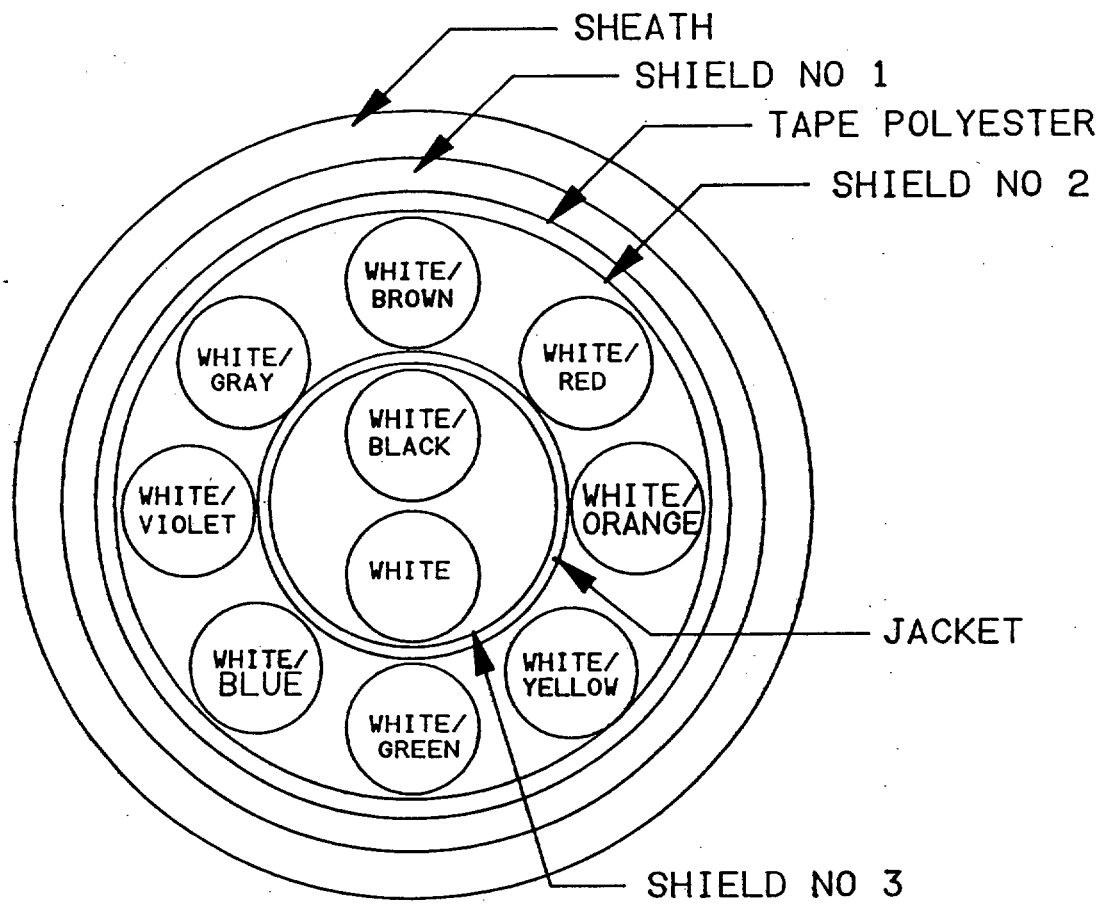


FIGURE 1. Cable configuration.

* The complete part number requires a color code designator (see table I).

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DESIGN DATA:

Design and construction: See figure 1.

Number of conductors: Ten.

Conductor size and stranding: See table I.

Wire: In accordance with M16878/17BDB*.

Jacket undershield: Not applicable.

Shielding: Tinned copper.

Sheath: Polyvinyl chloride.

Temperature rating: 105°C.

REQUIREMENTS:

Visual and dimensional:

Components:

Basic wire: Tinned copper.

Dimension: See table I.

Primary insulation:

Material: Polyvinyl chloride.

Wall thickness: .0096 inch ±.0025.

Colors:

White.

White with black tracer.

White with brown tracer.

White with red tracer.

White with orange tracer.

White with yellow tracer.

White with green tracer.

White with blue tracer.

White with violet tracer.

White with gray tracer.

Spark test voltage: 3,400 V rms.

Inspection test voltage: 2,000 V rms.

Impulse dielectric test: 8,000 V peak.

Jacket over insulation: Polyamide.

Wall thickness: .003 inch ±.001.

Color: Natural.

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Cabling:

Core:

Number of conductors: Two (one twisted pair).

Pitch or lay: 1.50 inches nominal.

Shield: Number 3.

Material: Tinned copper.

Gauge: 36 AWG.

Braid angle: 20 to 40 degrees.

First layer:

Number of conductors: Eight.

Pitch or lay: 3.00 inches nominal.

Shield: Number 2.

Material: Tinned copper.

Gauge: 36 AWG.

Braid angle: 20 to 40 degrees.

Tape: Over shield number 2.

Material: Polyester.

Workmanship: In accordance with MIL-C-27072.

Sheath:

Material: Polyvinyl chloride.

Wall thickness: .040 inch nominal.

Tensile strength: In accordance with table V of MIL-C-27072.

Elongation: In accordance with table V of MIL-C-27072.

Tension set: Not applicable.

Tear strength: Not applicable.

Accelerated aging: In accordance with table V of MIL-C-27072.

Oil resistance: Not applicable.

Color: See table IV of MIL-C-27072.

Finished cable:

Diameter: In accordance with table I.

Conductor resistance: In accordance with MIL-W-16878/17.

Dielectric strength: In accordance with MIL-W-16878/17.

Cold bend: Mandrel size in accordance with table XIII of MIL-C-27072.

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Ozone: Mandrel size in accordance with table XIV of MIL-C-27072.

Marking durability: In accordance with MIL-C-27072.

Heat stability: Not applicable.

TABLE I. Construction data.

Part number M27072/76	Conductor				Finished cable diameter (in.) ±10%	Basic wire M16878/17
	Wire size (AWG)	Number of strands	Strand size (AWG)	Nominal diameter (in.)		
SACA*	26	7	34	.041	.297	BDB***

Asterisks (*) shall be replaced by the appropriate color designator.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Air Force - 99
DLA - ES, IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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